

Problem A. Sereja and Dima

Time Limit 1000 ms

Mem Limit 262144 kB

Input File stdin

Output File stdout

[Problem Link](#)

Sereja and Dima play a game. The rules of the game are very simple. The players have n cards in a row. Each card contains a number, all numbers on the cards are distinct. The players take turns, Sereja moves first. During his turn a player can take one card: either the leftmost card in a row, or the rightmost one. The game ends when there is no more cards. The player who has the maximum sum of numbers on his cards by the end of the game, wins.

Sereja and Dima are being greedy. Each of them chooses the card with the larger number during his move.

Inna is a friend of Sereja and Dima. She knows which strategy the guys are using, so she wants to determine the final score, given the initial state of the game. Help her.

Input

The first line contains integer n ($1 \leq n \leq 1000$) — the number of cards on the table. The second line contains space-separated numbers on the cards from left to right. The numbers on the cards are distinct integers from 1 to 1000.

Output

On a single line, print two integers. The first number is the number of Sereja's points at the end of the game, the second number is the number of Dima's points at the end of the game.

Examples

Input	Output
4 4 1 2 10	12 5

Input	Output
7 1 2 3 4 5 6 7	16 12

Note

In the first sample Sereja will take cards with numbers **10** and **2**, so Sereja's sum is **12**.
Dima will take cards with numbers **4** and **1**, so Dima's sum is **5**.

Problem B. Make it White

Time Limit 2000 ms

Mem Limit 262144 kB

[Problem Link](#)

You have a horizontal strip of n cells. Each cell is either white or black.

You can choose a **continuous** segment of cells once and paint them all white. After this action, all the black cells in this segment will become white, and the white ones will remain white.

What is the minimum length of the segment that needs to be painted white in order for all n cells to become white?

Input

The first line of the input contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases. The descriptions of the test cases follow.

The first line of each test case contains a single integer n ($1 \leq n \leq 10$) — the length of the strip.

The second line of each test case contains a string s , consisting of n characters, each of which is either 'W' or 'B'. The symbol 'W' denotes a white cell, and 'B' — a black one. It is guaranteed that at least one cell of the given strip is black.

Output

For each test case, output a single number — the minimum length of a **continuous** segment of cells that needs to be painted white in order for the **entire** strip to become white.

Examples

Input	Output
8	4
6	1
WBBWBW	1
1	2
B	4
2	6
WB	4
3	7
BBW	
4	
BWWB	
6	
BWBWB	
6	
WWBBWB	
9	
WBWBWWBW	

Note

In the first test case of the example for the strip "WBBWBW", the minimum length of the segment to be repainted white is 4. It is necessary to repaint to white the segment from the 2-nd to the 5-th cell (the cells are numbered from 1 from left to right).

Problem C. Indian Summer

Time Limit 2000 ms

Mem Limit 262144 kB

Input File stdin

Output File stdout

[Problem Link](#)

Indian summer is such a beautiful time of the year! A girl named Alyona is walking in the forest and picking a bouquet from fallen leaves. Alyona is very choosy — she doesn't take a leaf if it matches the color and the species of the tree of one of the leaves she already has. Find out how many leaves Alyona has picked.

Input

The first line contains an integer n ($1 \leq n \leq 100$) — the number of leaves Alyona has found. The next n lines contain the leaves' descriptions. Each leaf is characterized by the species of the tree it has fallen from and by the color. The species of the trees and colors are given in names, consisting of no more than 10 lowercase Latin letters. A name can not be an empty string. The species of a tree and the color are given in each line separated by a space.

Output

Output the single number — the number of Alyona's leaves.

Examples

Input	Output
5 birch yellow maple red birch yellow maple yellow maple green	4

Input	Output
3 oak yellow oak yellow oak yellow	1

Problem D. Misha and Changing Handles

Time Limit 1000 ms

Mem Limit 262144 kB

Input File stdin

Output File stdout

[Problem Link](#)

Misha hacked the Codeforces site. Then he decided to let all the users change their handles. A user can now change his handle any number of times. But each new handle must not be equal to any handle that is already used or that was used at some point.

Misha has a list of handle change requests. After completing the requests he wants to understand the relation between the original and the new handles of the users. Help him to do that.

Input

The first line contains integer q ($1 \leq q \leq 1000$), the number of handle change requests.

Next q lines contain the descriptions of the requests, one per line.

Each query consists of two non-empty strings *old* and *new*, separated by a space. The strings consist of lowercase and uppercase Latin letters and digits. Strings *old* and *new* are distinct. The lengths of the strings do not exceed 20.

The requests are given chronologically. In other words, by the moment of a query there is a single person with handle *old*, and handle *new* is not used and has not been used by anyone.

Output

In the first line output the integer n — the number of users that changed their handles at least once.

In the next n lines print the mapping between the old and the new handles of the users. Each of them must contain two strings, *old* and *new*, separated by a space, meaning that

before the user had handle *old*, and after all the requests are completed, his handle is *new*. You may output lines in any order.

Each user who changes the handle must occur exactly once in this description.

Examples

Input	Output
5 Misha ILoveCodeforces Vasya Petrov Petrov VasyaPetrov123 ILoveCodeforces MikeMirzayanov Petya Ivanov	3 Petya Ivanov Misha MikeMirzayanov Vasya VasyaPetrov123

Problem E. Word Game

Time Limit 1000 ms

Mem Limit 262144 kB

[Problem Link](#)

Three guys play a game: first, each person writes down n distinct words of length 3. Then, they total up the number of points as follows:

- if a word was written by one person — that person gets 3 points,
- if a word was written by two people — each of the two gets 1 point,
- if a word was written by all — nobody gets any points.

In the end, how many points does each player have?

Input

The input consists of multiple test cases. The first line contains an integer t ($1 \leq t \leq 100$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains an integer n ($1 \leq n \leq 1000$) — the number of words written by each person.

The following three lines each contain n **distinct** strings — the words written by each person. Each string consists of 3 lowercase English characters.

Output

For each test case, output three space-separated integers — the number of points each of the three guys earned. You should output the answers in the same order as the input; the i -th integer should be the number of points earned by the i -th guy.

Examples

Input	Output
3 1 abc def abc 3 orz for qaq qaq orz for cod for ces 5 iat roc hem ica lly bac ter iol ogi sts bac roc lly iol iat	1 3 1 2 2 6 9 11 5

Note

In the first test case:

- The word **abc** was written by the first and third guys — they each get 1 point.
- The word **def** was written by the second guy only — he gets 3 points.

Problem F. YetnotherrokenKeoard

Time Limit 1000 ms

Mem Limit 262144 kB

[Problem Link](#)

Polycarp has a problem — his laptop keyboard is broken.

Now, when he presses the 'b' key, it acts like an unusual backspace: it deletes the last (rightmost) lowercase letter in the typed string. If there are no lowercase letters in the typed string, then the press is completely ignored.

Similarly, when he presses the 'B' key, it deletes the last (rightmost) uppercase letter in the typed string. If there are no uppercase letters in the typed string, then the press is completely ignored.

In both cases, the letters 'b' and 'B' are not added to the typed string when these keys are pressed.

Consider an example where the sequence of key presses was "ARaBbbiTBaby". In this case, the typed string will change as follows: "" $\xrightarrow{A}$ "A" $\xrightarrow{R}$ "AR" $\xrightarrow{a}$ "ARa" $\xrightarrow{B}$ "Aa" $\xrightarrow{b}$ "A" $\xrightarrow{b}$ "A" $\xrightarrow{i}$ "Ai" $\xrightarrow{t}$ "Ait" $\xrightarrow{B}$ "it" $\xrightarrow{a}$ "ita" $\xrightarrow{b}$ "it" $\xrightarrow{y}$ "ity".

Given a sequence of pressed keys, output the typed string after processing all key presses.

Input

The first line of the input data contains an integer t ($1 \leq t \leq 1000$), the number of test cases in the test.

The following contains t non-empty lines, which consist of lowercase and uppercase letters of the Latin alphabet.

It is guaranteed that each line contains at least one letter and the sum of the lengths of the lines does not exceed 10^6 .

Output

For each test case, output the result of processing the key presses on a separate line. If the typed string is empty, then output an empty line.

Examples

Input	Output
12 ARaBbbitBaby YetAnotherBrokenKeyboard Bubble Improbable abbreviable BbBB BusyasaBeeinaBedofBloomingBlossoms CoDEBARbIES codeforces bobebobbes b TheBBlackbboard	ity YetnotherrokenKeoard le Imprle revile usyasaeinaedofloominglossoms CDARIES codeforces es helaoard

Problem G. Registration System

Time Limit 5000 ms

Mem Limit 65536 kB

Input File stdin

Output File stdout

[Problem Link](#)

A new e-mail service "Berlandesk" is going to be opened in Berland in the near future. The site administration wants to launch their project as soon as possible, that's why they ask you to help. You're suggested to implement the prototype of site registration system. The system should work on the following principle.

Each time a new user wants to register, he sends to the system a request with his name. If such a name does not exist in the system database, it is inserted into the database, and the user gets the response OK, confirming the successful registration. If the name already exists in the system database, the system makes up a new user name, sends it to the user as a prompt and *also inserts the prompt into the database*. The new name is formed by the following rule. Numbers, starting with 1, are appended one after another to name (name1, name2, ...), among these numbers the least i is found so that name i does not yet exist in the database.

Input

The first line contains number n ($1 \leq n \leq 10^5$). The following n lines contain the requests to the system. Each request is a non-empty line, and consists of not more than 32 characters, which are all lowercase Latin letters.

Output

Print n lines, which are system responses to the requests: OK in case of successful registration, or a prompt with a new name, if the requested name is already taken.

Examples

Input	Output
4 abacaba acaba abacaba acab	OK OK abacaba1 OK

Input	Output
6 first first second second third third	OK first1 OK second1 OK third1

Problem H. Line

Time Limit 2000 ms

Mem Limit 262144 kB

[Problem Link](#)

There are n people in a horizontal line, each looking either to the left or the right. Each person counts the number of people in the direction they are looking. The **value** of the line is the sum of each person's count.

For example, in the arrangement LRRL, where L stands for a person looking left and R stands for a person looking right, the counts for each person are $[0, 3, 2, 3]$, and the value is $0 + 3 + 2 + 3 = 8$.

You are given the initial arrangement of people in the line. For each k from 1 to n , determine the maximum value of the line if you can change the direction of **at most** k people.

Input

The input consists of multiple test cases. The first line contains an integer t ($1 \leq t \leq 100$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains an integer n ($1 \leq n \leq 2 \cdot 10^5$) — the length of the line.

The following line contains a string consisting of n characters, each of which is either L or R, representing a person facing left or right, respectively — the description of the line.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Please note that the answer for some test cases won't fit into 32-bit integer type, so you should use at least 64-bit integer type in your programming language (like `long long` for C++).

Output

For each test case, output n space-separated non-negative integers — the maximum value of the line if you can change the direction of **at most** k people for each k from 1 to n ,

inclusive.

Examples

Input	Output
6	3 5 5
3	16 16 16 16 16
LLR	0
5	86 95 98 101 102 102 102 102 102 102 102
LRRL	102
1	29 38 45 52 57 62 65 68 69 70
L	44 50 54 56 56 56 56 56 56
12	
LRRLLLRLRL	
10	
LLLLRRRR	
9	
LRLRLRL	

Note

In the first test case:

- $k = 1$: change the direction of 1 person to make the line RLR. The total value is $2 + 1 + 0 = 3$.
- $k = 2$: change the direction of 2 people to make the line RLL. The total value is $2 + 1 + 2 = 5$.
- $k = 3$: change the direction of 2 people to make the line RLL. The total value is $2 + 1 + 2 = 5$. Note that you have to change the direction of **at most** k people.

In the second test case, it is optimal to only change the direction of the first person for all k from 1 to 5 (that is, make the line RRRL).

Problem I. Chat Order

Time Limit 3000 ms

Mem Limit 262144 kB

[Problem Link](#)

Polycarp is a big lover of killing time in social networks. A page with a chatlist in his favourite network is made so that when a message is sent to some friend, his friend's chat rises to the very top of the page. The relative order of the other chats doesn't change. If there was no chat with this friend before, then a new chat is simply inserted to the top of the list.

Assuming that the chat list is initially empty, given the sequence of Polycarpus' messages make a list of chats after all of his messages are processed. Assume that no friend wrote any message to Polycarpus.

Input

The first line contains integer n ($1 \leq n \leq 200\,000$) — the number of Polycarpus' messages. Next n lines enlist the message recipients in the order in which the messages were sent. The name of each participant is a non-empty sequence of lowercase English letters of length at most 10.

Output

Print all the recipients to who Polycarp talked to in the order of chats with them, from top to bottom.

Examples

Input	Output
4 alex ivan roman ivan	ivan roman alex

Input	Output
8 alina maria ekaterina darya darya ekaterina maria alina	alina maria ekaterina darya

Note

In the first test case Polycarpus first writes to friend by name "alex", and the list looks as follows:

1. alex

Then Polycarpus writes to friend by name "ivan" and the list looks as follows:

1. ivan
2. alex

Polycarpus writes the third message to friend by name "roman" and the list looks as follows:

1. roman
2. ivan
3. alex

Polycarpus writes the fourth message to friend by name "ivan", to who he has already sent a message, so the list of chats changes as follows:

1. ivan
2. roman
3. alex

Problem J. Tom Riddle's Diary

Time Limit 2000 ms

Mem Limit 262144 kB

[Problem Link](#)

Harry Potter is on a mission to destroy You-Know-Who's Horcruxes. The first Horcrux that he encountered in the Chamber of Secrets is Tom Riddle's diary. The diary was with Ginny and it forced her to open the Chamber of Secrets. Harry wants to know the different people who had ever possessed the diary to make sure they are not under its influence.

He has names of n people who possessed the diary in order. You need to tell, for each person, if he/she possessed the diary at some point before or not.

Formally, for a name S_i in the i -th line, output "YES" (without quotes) if there exists an index j such that $S_i = S_j$ and $j < i$, otherwise, output "NO" (without quotes).

Input

First line of input contains an integer n ($1 \leq n \leq 100$) — the number of names in the list.

Next n lines each contain a string S_i , consisting of lowercase English letters. The length of each string is between 1 and 100.

Output

Output n lines each containing either "YES" or "NO" (without quotes), depending on whether this string was already present in the stream or not.

You can print each letter in any case (upper or lower).

Examples

Input	Output
6 tom lucius ginny harry ginny harry	NO NO NO NO YES YES YES

Input	Output
3 a a a	NO YES YES YES

Note

In test case 1, for $i = 5$ there exists $j = 3$ such that $s_i = s_j$ and $j < i$, which means that answer for $i = 5$ is "YES".

Problem K. Strong Password

Time Limit 2000 ms

Mem Limit 524288 kB

[Problem Link](#)

Monocarp's current password on Codeforces is a string s , consisting of lowercase Latin letters. Monocarp thinks that his current password is too weak, so he wants to **insert exactly one lowercase Latin letter** into the password to make it stronger. Monocarp can choose any letter and insert it anywhere, even before the first character or after the last character.

Monocarp thinks that the password's strength is proportional to the time it takes him to type the password. The time it takes to type the password is calculated as follows:

- the time to type the first character is 2 seconds;
- for each character other than the first, the time it takes to type it is 1 second if it is the same as the previous character, or 2 seconds otherwise.

For example, the time it takes to type the password `abacaba` is 14; the time it takes to type the password `a` is 2; the time it takes to type the password `aaabacc` is 11.

You have to help Monocarp — insert a lowercase Latin letter into his password so that the resulting password takes the maximum possible amount of time to type.

Input

The first line contains one integer t ($1 \leq t \leq 1000$) — the number of test cases.

Each test case consists of one line containing the string s ($1 \leq |s| \leq 10$), consisting of lowercase Latin letters.

Output

For each test case, print one line containing the new password — a string which can be obtained from s by inserting one lowercase Latin letter. The string you print should have

the maximum possible required time to type it. If there are multiple answers, print any of them.

Examples

Input	Output
4 a aaa abb password	wa aada abcb pastsword

Problem L. Symmetric Encoding

Time Limit 2000 ms

Mem Limit 262144 kB

[Problem Link](#)

Polycarp has a string s , which consists of lowercase Latin letters. He encodes this string using the following algorithm:

- first, he constructs a new auxiliary string r , which consists of all distinct letters of the string s , written in alphabetical order;
- then the encoding happens as follows: each character in the string s is replaced by its symmetric character from the string r (the first character of the string r will be replaced by the last, the second by the second from the end, and so on).

For example, encoding the string $s = \text{"codeforces"}$ happens as follows:

- the string r is obtained as "cdefors" ;
- the first character $s_1 = \text{'c'}$ is replaced by 's' ;
- the second character $s_2 = \text{'o'}$ is replaced by 'e' ;
- the third character $s_3 = \text{'d'}$ is replaced by 'r' ;
- ...
- the last character $s_{10} = \text{'s'}$ is replaced by 'c' .



The string r and replacements for $s = \text{"codeforces"}$.

Thus, the result of encoding the string $s = \text{"codeforces"}$ is the string "serofedsoc" .

Write a program that performs decoding — that is, restores the original string s from the encoding result.

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains a single integer n ($1 \leq n \leq 2 \cdot 10^5$) — the length of the string b .

The second line of each test case contains a string b of length n , consisting of lowercase Latin letters — the result of encoding the original string s .

It is guaranteed that the sum of the values of n over all test cases in the test does not exceed $2 \cdot 10^5$.

Output

For each test case, output the string s from which the encoding result b was obtained.

Examples

Input	Output
5 10 serofedsoc 3 ttf 9 tlrhgmaoi 1 w 15 hnnldedmnhlttin	codeforces fft algorithm w meetinthemiddle

Problem M. Lady Bug

Time Limit 1000 ms

Mem Limit 262144 kB

[Problem Link](#)

As soon as Dasha Purova crossed the border of France, the villain Markaron kidnapped her and placed her in a prison under his large castle. Fortunately, the wonderful Lady Bug, upon hearing the news about Dasha, immediately ran to save her in Markaron's castle. However, to get there, she needs to crack a complex password.

The password consists of two bit strings a and b , each of which has a length of n . In one operation, Lady Bug can choose any index $2 \leq i \leq n$ and perform one of the following actions:

1. $\text{swap}(a_i, b_{i-1})$ (swap the values of a_i and b_{i-1}), or
2. $\text{swap}(b_i, a_{i-1})$ (swap the values of b_i and a_{i-1}).

Lady Bug can perform any number of operations. The password is considered cracked if she can ensure that the first string consists only of zeros. Help her understand whether or not she will be able to save the unfortunate Dasha.

Input

Each test consists of several test cases. The first line of the input data contains one integer t ($1 \leq t \leq 10^4$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains one integer n ($2 \leq n \leq 2 \cdot 10^5$) — the length of the bit strings of the password.

The next two lines contain the bit strings of length n , a and b , which represent the password. Each of the strings contains only the characters 0 and '1'.

It is guaranteed that the sum of n across all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output "YES" if Lady Bug can crack the password after any number of operations; otherwise, output "NO".

You can output each letter in any case (lowercase or uppercase). For example, the strings "yEs", "yes", "Yes", and "YES" will be accepted as a positive answer.

Examples

Input	Output
4 3 000 000 6 010001 010111 5 10000 01010 2 11 00	YES YES NO YES

Note

In the first test case, the string a immediately consists only of zeros.

In the second test case, a possible sequence of operations is:

1. $\text{swap}(a_2, b_1)$

0**1**0001

010111

2. $\text{swap}(b_5, a_4)$

000**0**01

1101**1**1

3. $\text{swap}(a_4, b_3)$

000**1**01

11**0**101

4. $\text{swap}(a_5, b_4)$

00000**1**

111101

Problem N. Remove Prefix

Time Limit 2000 ms

Mem Limit 262144 kB

[Problem Link](#)

Polycarp was presented with some sequence of integers a of length n ($1 \leq a_i \leq n$). A sequence can make Polycarp happy only if it consists of **different** numbers (i.e. distinct numbers).

In order to make his sequence like this, Polycarp is going to make some (possibly zero) number of moves.

In one move, he can:

- remove the first (leftmost) element of the sequence.

For example, in one move, the sequence $[3, 1, 4, 3]$ will produce the sequence $[1, 4, 3]$, which consists of different numbers.

Determine the minimum number of moves he needs to make so that in the remaining sequence all elements are different. In other words, find the length of the smallest prefix of the given sequence a , after removing which all values in the sequence will be unique.

Input

The first line of the input contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

Each test case consists of two lines.

The first line contains an integer n ($1 \leq n \leq 2 \cdot 10^5$) — the length of the given sequence a .

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq n$) — elements of the given sequence a .

It is guaranteed that the sum of n values over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case print your answer on a separate line — the minimum number of elements that must be removed from the beginning of the sequence so that all remaining elements are different.

Examples

Input	Output
5	1
4	4
3 1 4 3	0
5	0
1 1 1 1 1	5
1	
1	
6	
6 5 4 3 2 1	
7	
1 2 1 7 1 2 1	

Note

The following are the sequences that will remain after the removal of prefixes:

- [1, 4, 3];
- [1];
- [1];
- [6, 5, 4, 3, 2, 1];
- [2, 1].

It is easy to see that all the remaining sequences contain only distinct elements. In each test case, the shortest matching prefix was removed.

Problem O. Two-gram

Time Limit 1000 ms

Mem Limit 262144 kB

[Problem Link](#)

Two-gram is an ordered pair (i.e. string of length two) of capital Latin letters. For example, "AZ", "AA", "ZA" — three distinct two-grams.

You are given a string S consisting of n capital Latin letters. Your task is to find **any** two-gram contained in the given string **as a substring** (i.e. two consecutive characters of the string) maximal number of times. For example, for string $S = \text{"BBAABBBBA"}$ the answer is two-gram "BB", which contained in S three times. In other words, find any most frequent two-gram.

Note that occurrences of the two-gram can overlap with each other.

Input

The first line of the input contains integer number n ($2 \leq n \leq 100$) — the length of string S . The second line of the input contains the string S consisting of n capital Latin letters.

Output

Print the only line containing exactly two capital Latin letters — **any** two-gram contained in the given string S **as a substring** (i.e. two consecutive characters of the string) maximal number of times.

Examples

Input	Output
7 ABACABA	AB

Input	Output
5 ZZZAA	ZZ

Note

In the first example "BA" is also valid answer.

In the second example the only two-gram "ZZ" can be printed because it contained in the string "ZZZAA" two times.

Problem P. Permutation Minimization by Deque

Time Limit 2000 ms

Mem Limit 262144 kB

[Problem Link](#)

In fact, the problems E1 and E2 do not have much in common. You should probably think of them as two separate problems.

A permutation p of size n is given. A *permutation* of size n is an array of size n in which each integer from 1 to n occurs exactly once. For example, $[1, 4, 3, 2]$ and $[4, 2, 1, 3]$ are correct permutations while $[1, 2, 4]$ and $[1, 2, 2]$ are not.

Let us consider an empty [deque](#) (double-ended queue). A deque is a data structure that supports adding elements to both the beginning and the end. So, if there are elements $[1, 5, 2]$ currently in the deque, adding an element 4 to the beginning will produce the sequence $[4, 1, 5, 2]$, and adding same element to the end will produce $[1, 5, 2, 4]$.

The elements of the permutation are sequentially added to the initially empty deque, starting with p_1 and finishing with p_n . Before adding each element to the deque, you may choose whether to add it to the beginning or the end.

For example, if we consider a permutation $p = [3, 1, 2, 4]$, one of the possible sequences of actions looks like this:

1.	add 3 to the end of the deque:	deque has a sequence $[3]$ in it;
2.	add 1 to the beginning of the deque:	deque has a sequence $[1, 3]$ in it;
3.	add 2 to the end of the deque:	deque has a sequence $[1, 3, 2]$ in it;
4.	add 4 to the end of the deque:	deque has a sequence $[1, 3, 2, 4]$ in it;

Find the lexicographically smallest possible sequence of elements in the deque after the entire permutation has been processed.

A sequence $[x_1, x_2, \dots, x_n]$ is *lexicographically smaller* than the sequence $[y_1, y_2, \dots, y_n]$ if there exists such $i \leq n$ that $x_1 = y_1, x_2 = y_2, \dots, x_{i-1} = y_{i-1}$ and $x_i < y_i$. In other words, if the sequences x and y have some (possibly empty) matching prefix, and the next element of the sequence x is strictly smaller than the corresponding element of the sequence y . For example, the sequence $[1, 3, 2, 4]$ is smaller than the sequence $[1, 3, 4, 2]$

because after the two matching elements $[1, 3]$ in the start the first sequence has an element 2 which is smaller than the corresponding element 4 in the second sequence.

Input

The first line contains an integer t ($1 \leq t \leq 1000$) — the number of test cases.

The next $2t$ lines contain descriptions of the test cases.

The first line of each test case description contains an integer n ($1 \leq n \leq 2 \cdot 10^5$) — permutation size. The second line of the description contains n space-separated integers p_i ($1 \leq p_i \leq n$; all p_i are all unique) — elements of the permutation.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

Print t lines, each line containing the answer to the corresponding test case. The answer to a test case should contain n space-separated integer numbers — the elements of the lexicographically smallest permutation that is possible to find in the deque after executing the described algorithm.

Examples

Input	Output
5	1 3 2 4
4	1 2 3
3 1 2 4	1 3 2
3	1 2
3 2 1	1 2
3	
3 1 2	
2	
1 2	
2	
2 1	

Note

One of the ways to get a lexicographically smallest permutation $[1, 3, 2, 4]$ from the

permutation $[3, 1, 2, 4]$ (the first sample test case) is described in the problem statement.

Problem Q. Palindrome Reorder

Time Limit 1000 ms

Mem Limit 524288 kB

[Problem Link](#)

Given a string, your task is to reorder its letters in such a way that it becomes a palindrome (i.e., it reads the same forwards and backwards).

Input

The only input line has a string of length n consisting of characters A–Z.

Output

Print a palindrome consisting of the characters of the original string. You may print any valid solution. If there are no solutions, print "NO SOLUTION".

Constraints

- $1 \leq n \leq 10^6$

Example

Input	Output
AAAACACBA	AACABACAA

Problem R. FIGUREFUL

Time Limit	1000 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

[Problem Link](#)

Don and his friends had some sort of code word to describe the girls who pass towards their class.

- Anitha passed - killer told.. “Dude, 5 2 is going”
- Chitra passed - Jumbo told.. “Dude, 6 4 is awesome”
- Divya passed - Baabi told.. “Machi 1 6 is my dream girl”

Girls couldn't understand those code words. To crack the code word, girls decide to record all the code words and the girls' name whenever they call.

The silencer in the class watched all these and got angry. He mugged up all the code words, went to teacher and complained about the naughty boys and naughty girls.

- **Teacher:** Who is the girl with code word 1 6?
- **Silencer:** Divya
- **Teacher:** Who is the girl with code word 5 2?
- **Silencer:** Anitha
- **Teacher:** What is the code word for me?
- **Silencer:** (puzzled)

Now you have to pretend that you are the Silencer to solve this problem. Record the code words and answer the questions the teacher asks you. (She may ask the same question any number of times.)

Note: There can be any number of code words for a same girl. If there exist more than one girl for a same code word, then that code word belongs to the latest girl.

Input

The first line contains an integer n , the number of figures. Then in each of the next n lines there will be 2 integers and a figure name.

After that there will be an integer t , the number of test cases. For each test case, there will be two integers representing the code word. Print the figure's name corresponding to the code word.

Output

For each test case output the figure name.

Constraints

$$1 \leq n \leq 100000$$

$$1 \leq t \leq 100000$$

$$-1000000 \leq \text{code1}, \text{code2} \leq 1000000$$

The figure's name will not exceed 20 characters.

Example

Input	Output
6 5 2 Anitha 6 4 Yamuna 6 4 Chitra 1 6 Divya -2 0 Teacher 7 4 Anitha 2 -2 0 6 4	Teacher Chitra

Hint: Use map.

Problem S. Ada and Queue

Time Limit	3500 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

[Problem Link](#)

Ada the Ladybug has many things to do. She puts them into her queue. Anyway she is very indecisive, so sometime she uses the top, sometime the back and sometime she decides to reverses it.

Input

The first line consists of $1 \leq Q \leq 10^6$, number of queries. Each of them contains one of following commands

back - Print number from back and then erase it

front - Print number from front and then erase it

reverse - Reverses all elements in queue

push_back N - Add element N to back

toFront N - Put element N to front

All numbers will be $0 \leq N \leq 100$

Output

For each back/front query print appropriate number.

If you would get this type of query and the queue would be empty, print "No job for Ada?" instead.

Example

Input	Output
15 toFront 93 front back reverse back reverse toFront 80 push_back 53 push_back 50 front front reverse push_back 66 reverse front	93 No job for Ada? No job for Ada? 80 53 66

Problem T. Backspace

Time Limit 1000 ms

Mem Limit 1048576 kB

OS	Linux
----	-------

[Problem Link](#)

Right before Forritunarkeppni Framhaldsskólanna started, Bjarki decided to update his computer. He didn't notice anything off until he started writing code in his favourite editor Bim (Bjarki IMproved).

Usually when he uses this editor

and presses *backspace* one letter to the left of the cursor is erased. But after the update it instead writes the letter <. He has tried all of his editors, Bmacs, Neobim, bjedit, NoteBjad++, Subjark Text, but they all seem to have this same issue. He doesn't have the time to scour the internet to find a solution to this problem, so he decides to take matters into his own hands and just solve this.

Help Bjarki write a program that takes as input a string he wrote and prints the string as he intended to write it.

Input

The first and only line of input contains a string S of length N containing only lower case English letters and the symbol $<$.

Output

Print the string as Bjarki intended to write it. That is to say, if the string is printed letter by letter, `<` denotes the removal of the last printed letter as if it were a *backspace*.

Explanation of Sample Inputs

In the first sample Bjarki starts by writing a a which he then erases, then writes b and c but then erases the c. The output is thus b. In the next sample Bjarki writes foss but then erases the last two letters with <<, the output at this point is thus fo. He then adds to this

```
bla.cpp (/tmp) - VIM
1 #include <bits/stdc++.h>
2 using namespace std;
3
4 int main() {
5     int n;
6     cin >> n;
7     vector<int> v(n);
8     for (int i = 0; i < n; i++) {
9         cin >> v[i];
10    }
11    sort(v.begin(), v.end());
12    int ans = 0;
13    for (int i = 0; i < n; i++) {
14        if (v[i] % 2 == 0) {
15            ans += v[i];
16        }
17    }
18    cout << ans << endl;
19    return 0;
20 }
```

rritun and the output is thus forritun. In the last sample he writes a and erases it immediately afterwards twice. Then finally he writes aa and erases it with <<.

Scoring

The solution will be tested on differently hard input data and the data is divided into groups as shown in the table below. The solution will then be scored according to how many groups are solved.

Group	Points	Constraints
1	10	$1 \leq N \leq 100$, The string S only contains the letter a and the symbol <
2	10	$1 \leq N \leq 100$, The string S does not contain two consecutive <
3	40	$1 \leq N \leq 100$
4	40	$1 \leq N \leq 10^6$

Sample 1

Input	Output
a<bc<	b

Sample 2

Input	Output
foss<<rritun	forritun

Sample 3

Input	Output
a<a<a<aa<<	

Problem U. pushpush

Time Limit 2000 ms

Mem Limit 262144 kB

[Problem Link](#)

Problem Statement

You are given an integer sequence of length n , $a_1, \dots, a_n$. Let us consider performing the following n operations on an empty sequence b .

The i -th operation is as follows:

1. Append a_i to the end of b .
2. Reverse the order of the elements in b .

Find the sequence b obtained after these n operations.

Constraints

- $1 \leq n \leq 2 \times 10^5$
- $0 \leq a_i \leq 10^9$
- n and a_i are integers.

Input

Input is given from Standard Input in the following format:

```
 $n$   
 $a_1$   $a_2$  ...  $a_n$ 
```

Output

Print n integers in a line with spaces in between. The i -th integer should be b_i .

Sample 1

Input	Output
4 1 2 3 4	4 2 1 3

- After step 1 of the first operation, b becomes: 1.
- After step 2 of the first operation, b becomes: 1.
- After step 1 of the second operation, b becomes: 1, 2.
- After step 2 of the second operation, b becomes: 2, 1.
- After step 1 of the third operation, b becomes: 2, 1, 3.
- After step 2 of the third operation, b becomes: 3, 1, 2.
- After step 1 of the fourth operation, b becomes: 3, 1, 2, 4.
- After step 2 of the fourth operation, b becomes: 4, 2, 1, 3.

Thus, the answer is 4 2 1 3.

Sample 2

Input	Output
3 1 2 3	3 1 2

As shown above in Sample Output 1, b becomes 3, 1, 2 after step 2 of the third operation. Thus, the answer is 3 1 2.

Sample 3

Input	Output
1 1000000000	1000000000

Sample 4

Input	Output
6 0 6 7 6 7 0	0 6 6 0 7 7